A Planning Proposal for a Bounded Phase I Feasibility Study of an Open-Source Coordination Framework for Local Agricultural Supply in Summit County, Ohio

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Technical Report Writing

Proposal Draft

April 24, 2026

I. Responsiveness to USDA NIFA SBIR Program Priorities

This proposal addresses USDA SBIR Topic 8.6, Rural and Community Development. It is written as a **planning proposal whose Phase I work takes the form of a bounded feasibility study**. The planning dimension concerns a proposed path for improving how local agricultural communities coordinate supply, planning, and market access. The feasibility dimension concerns whether that path is technically and economically credible enough to justify Phase II development.

The project is responsive to rural and community development priorities because it addresses a structural problem that affects the viability of small and independent farms as local economic actors. In rural and peri-rural communities, farms often remain operationally isolated even when nearby markets value freshness, local origin, and production practices associated with independently operated farms. The issue is not merely one of demand visibility. It is also one of fragmented information, inconsistent planning signals, weak interoperability across operators, and the absence of a coordination framework that allows local supply to be used as a coherent resource (Dorn, 2022; USDA ERS, 2026).

This proposal does not claim that a full production-ready system has already been validated. Instead, it seeks approval to conduct a bounded Phase I study that tests whether a realistic coordination model can reveal sourcing scenarios in which local farms improve their position while buyers remain at least non-worse-off.

II. Identification and Significance of the Problem or Opportunity

The problem addressed by this proposal is that local agriculture remains underutilized even where its products are socially valued and geographically adjacent to demand. Small and independent farms frequently operate in conditions where planning, sourcing, and sell-through

are uncertain. This uncertainty discourages reinvestment, limits reliable output, increases coordination friction, and prevents local agriculture from functioning as a stable resource for nearby buyers (Bielaczyc et al., 2020; Barry et al., 2025).

The project treats this as a **coordination failure** rather than a simple absence of consumer interest. Farms, buyers, and intermediary organizations often manage information through disconnected systems, informal communication, or manual workarounds. Under such conditions, local supply may exist but remain difficult to aggregate, compare, route, and evaluate against buyer requirements. As a result, local participation becomes harder to scale into regular procurement relationships (Bielaczyc et al., 2020; Barry et al., 2025; Ge et al., 2019).

The significance of the problem is economic, technical, and community-facing. Economically, instability in local sell-through conditions can contribute to volatility in farm planning and reinvestment. Technically, the lack of interoperable data structures makes decentralized coordination difficult to test and harder to operationalize. At the community level, this reduces the ability of rural and local agricultural networks to function as durable, self-reinforcing systems of production and exchange. National evidence quantifies each dimension. Farm household income is substantially more volatile than off-farm income, as measured by the coefficient of variation of monthly net cash margin, making sell-through uncertainty a particularly consequential constraint on farm planning and reinvestment (Key et al., 2017). The structural weight of post-farm marketing functions further concentrates efficiency advantages in centralized supply chains, where most consumer food spending accrues after the farm gate—farmers received 15.9 cents of each dollar spent on domestically produced food in 2023 (USDA ERS, 2026). Compliance costs compound this disadvantage: regulatory costs under the FSMA Produce Rule range from 0.3 percent of annual produce sales for the largest farm operations to

6.8 percent for the smallest, and retailers commonly impose additional third-party audit requirements beyond what the regulation itself requires (Bovay et al., 2018; Minor et al., 2019).

For the purposes of this proposal, the practical problem is defined narrowly: **some portion of local farm underutilization may result from fragmented coordination between farms, buyers, and intermediary institutions rather than from an absolute absence of viable demand**. Phase I therefore asks whether a structured coordination framework can identify a feasible transition pathway under bounded conditions.

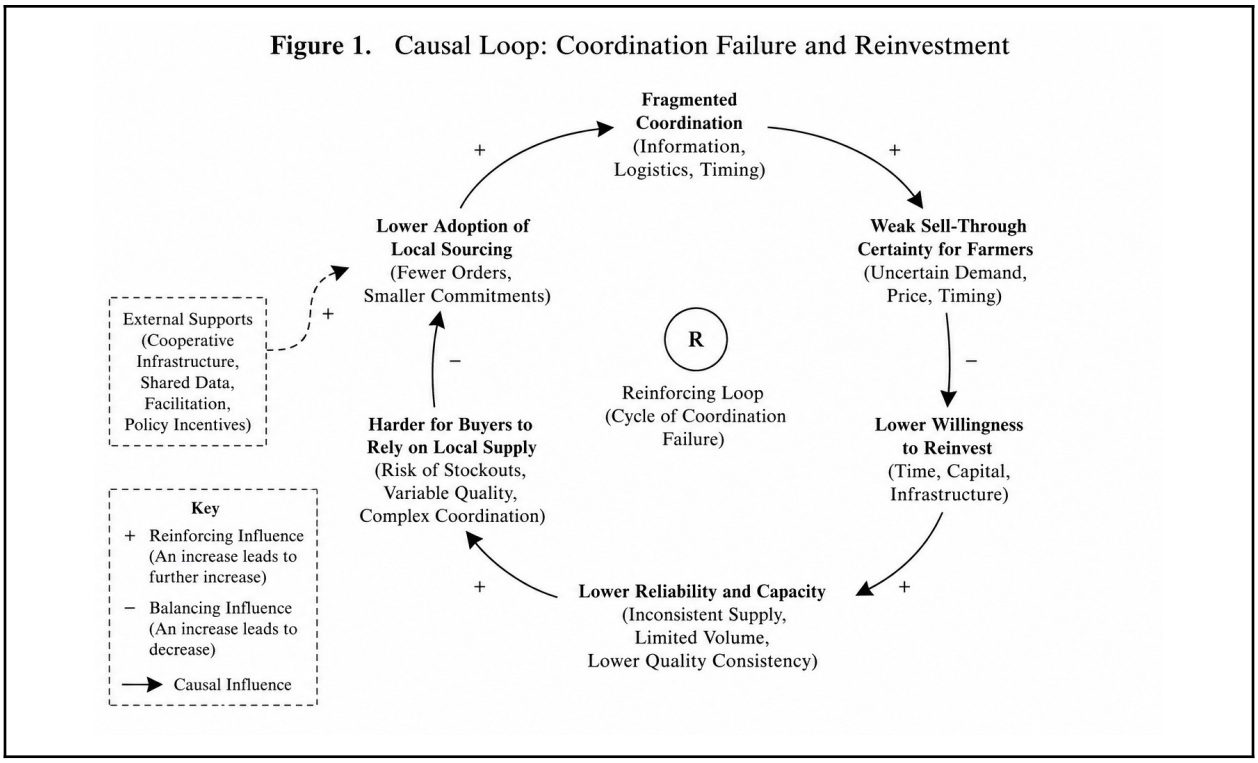
Target beneficiaries

The primary beneficiaries are small and independent farms whose operations are constrained by fragmented planning, uncertain sell-through, and weak coordination infrastructure. Secondary beneficiaries include local food hubs, grocers, and community-serving buyers that may benefit from greater visibility into nearby supply if that supply can be expressed and assessed more consistently. Rural communities benefit indirectly if farm viability, repeatable sourcing, and local reinvestment become more plausible under improved coordination conditions.

Socio-economic framing

The proposal begins from the view that increasingly it is inequitable to work with and in local agriculture, even though local agriculture remains a central organizing force in rural communities. The concern is not simply that farms produce less. It is that the conditions under which they plan, invest, and reach markets are unstable. This instability constrains farm profitability and suppresses broader local value realization.

Figure 1



Causal loop diagram of coordination failure, underutilization, and weak reinvestment.

III. Background and Rationale

Existing local and regional food systems show that there is no shortage of interest in models that improve coordination. Food hubs, marketplaces, recordkeeping systems, and proprietary farm-management tools each address part of the problem. However, current solutions often improve only one layer of the system at a time: transaction access, internal recordkeeping, compliance tracking, or isolated inventory visibility. They do not necessarily make multiple independent farms legible as a coordinated operational resource without relying on a central platform to own or govern the relationship (Bielaczyc et al., 2020; Barry et al., 2025).

The rationale for this proposal is therefore not that no tools exist, but that current tools do not sufficiently solve the problem of decentralized interoperability. Existing systems may help a farm manage itself, or help a buyer transact in a managed marketplace, but they often do not provide an open, transferable coordination layer that allows multiple participants to express

supply, constraints, and operational context in a form that can be recombined without central ownership of the network.

This proposal positions Fruitful Network Development's approach as a bounded attempt to test that missing layer. The project is not framed as a universal replacement for existing software. Nor is it framed as proof that all local agriculture can be made competitive through software alone. Instead, it is framed as a test of whether an open-source coordination framework can reduce some of the information and planning friction that currently prevents local supply from being evaluated and used more coherently.

Need for a new approach

Fruitful Network Development is not simply proposing another inventory application or marketplace. The intended innovation is a foundational coordination framework that allows farms to express operational and supply information from the same system they use to manage their own activity, while preserving local autonomy. The innovation is not in controlling the network. It is in making self-organization more technically possible.

At a practical level, this approach is intended to support farm-side planning, supply-side visibility, and buyer-side scenario evaluation without requiring a single central authority to act as the sole point of truth. This distinction matters because many local agricultural systems become dependent on the success or limitations of one intermediary. The proposed framework instead asks whether a shared data model and bounded coordination logic can help expose unrealized value across a local network.

End-user requirements

For this proposal, end-user requirements are defined narrowly. Farms need a way to represent current and expected output, capacity bands, harvest windows, location, and

operational constraints in a structured form. Buyers or intermediary purchasers need a way to express bounded demand profiles, service requirements, and tolerance conditions. Phase I does not need to solve every user workflow. It needs to solve enough of the representational and coordination problem to support a credible feasibility test.

Background limitations

National food-hub research confirms these structural patterns. The 2019 National Food Hub Survey identified balancing supply and demand as the sector's most persistent operational challenge, while meeting buyer pricing requirements was the most commonly cited barrier to expanding institutional sales (Bielaczyc et al., 2020). The 2025 survey found that growth expectations in the sector coexist with continuing constraints in price negotiation, capital access, infrastructure investment, and trucking and logistics management (Barry et al., 2025). Modeling work on fresh-produce aggregation hub networks further shows that economies of scale materially influence the optimal number, size, and placement of coordination nodes, confirming that local coordination must be tested as a cost-structure question rather than simply assumed to be viable (Ge et al., 2019). Perishability sharpens all of these challenges: the USDA Economic Research Service documents that total losses of fresh fruit reach 37 percent and fresh vegetables 34 percent at the retail and consumer levels alone, excluding unmeasured farm-level losses, which makes timing precision, handling quality, and coordination discipline economically consequential rather than incidental (USDA ERS, 2025). Consumer willingness to pay for local food is not a reliable substitute for operational feasibility: a meta-regression analysis found substantial heterogeneity and evidence of publication bias in the willingness-to-pay literature (Printezis et al., 2019), and a controlled field experiment found no price premium attributable to a generic local label under competitive conditions (Davidson et al., 2024). The operational case

for improved coordination must therefore rest on demonstrated cost-structure and service-level feasibility, not on assumed consumer-side price premiums.

Figure 2

Figure 2. Comparison of Existing Solutions Against Key Needs

Existing Solution Category	Internal Farm Management (Operations, Inventory, Planning)	Buyer Coordination (Discovery, Matching, Ordering, Scheduling)	Interoperability Across Actors (Open Data Exchange & Shared Standards)	Dependence on Central Authority (Platform or Data Control)	Relevance to Phase I Feasibility Study
Proprietary Farm Management Software *(e.g., operation management tools)*	● Strong	○ Weak	○ Weak (Proprietary formats, limited exchange)	● High	○ Low–Moderate (Partial fit; not built *for multi-actor coordination)*
Marketplace / Platform Solutions *(e.g., online marketplaces and aggregator platforms)*	○ Weak	● Strong	○ Weak–Moderate (Closed ecosystems; limited portability)	● High	○ Low–Moderate (Not designed for decentralized interoperability)
Food Hubs / Aggregators *(e.g., regional aggregation and distribution hubs)*	○ Weak	● Strong (with participating buyers)	◑ Moderate (Varies by hub; often custom systems)	◑ Moderate (Operational control by hub)	◑ Moderate (Helpful channel, but not a general interoperability layer)
Compliance / Record Keeping Tools *(e.g., food safety, traceability, audit management)*	○ Weak	○ Weak	◑ Moderate (Export capability varies)	● High	○ Low (Solves compliance narrowly; not built for coordination)

● Strong alignment ◑ Moderate alignment ○ Weak alignment

Note. This comparison reflects typical characteristics of solution categories. Individual products may vary.

Existing-solutions comparison chart.

IV. Relationship to Research and Phase II

Phase I is intended to answer a threshold question: **is there a realistic, bounded pathway by which local supply in the Summit County study area could be coordinated so that participating farms improve stability or expected gain while buyers remain at least non-worse-off?**

The answer to that question determines whether broader operational development is justified. If the Phase I study identifies a feasible window under realistic assumptions, Phase II would focus on operationalizing the minimum commitments and technical workflows needed to use the framework in recurring transactions. If Phase I fails to identify such a window, the project will still produce useful outcomes by clarifying which constraints remain binding and what would need to change before retesting feasibility.

Expected Phase I outputs

Phase I is expected to produce four core outputs:

1. a structured minimum data model for participating farms, buyers, and scenario variables;
2. a prototype coordination framework sufficient to normalize, compare, and process those inputs;
3. a bounded simulation environment that tests local sourcing scenarios under defined constraints;
4. a feasibility assessment identifying whether a two-sided feasible window exists and, if so, under what conditions.

Relationship to Phase II

Phase II would not begin from abstraction. It would begin from the Phase I outputs. If feasibility is supported, Phase II would focus on deployment-oriented tasks such as broader farm onboarding, live workflow refinement, recurring routing and aggregation logic, integration with more active user operations, and expansion of the basket or network size under validated conditions.

Phase II trigger

Phase II is justified only if Phase I identifies at least one credible scenario in which both sides of the market remain within acceptable bounds. A positive result would not mean the full business model is proven. It would mean that advancing to implementation is rational. A negative result would not invalidate the entire concept. It would mean the study did not find a feasible transition under the conditions tested.

V. Technical Objectives

The technical objective of Phase I is not to build a complete market platform. It is to determine whether an open-source coordination framework can support a credible feasibility test of localized agricultural sourcing under bounded conditions.

Objective 1: Define the minimum representational layer required for coordination

Phase I will define the minimum structured elements needed to represent farm-side and buyer-side conditions in a form that can be compared and evaluated across participants. This includes operational attributes such as current output bands, capacity bands, timing windows, and coordination-relevant constraints.

Output: minimum schema / structured data specification

Success condition: the representation is sufficient to parameterize the study without requiring universal standardization

Objective 2: Build the minimum prototype needed to run the feasibility study

Phase I will develop a prototype coordination framework that can ingest structured inputs, normalize them, apply bounded constraints, and generate interpretable outputs. The prototype exists to enable analysis, not to serve as proof of a finished commercial product.

Output: working prototype sufficient for scenario testing

Success condition: prototype can accept structured study inputs and produce scenario outputs without manual rewriting of the entire workflow

Objective 3: Parameterize a bounded local sourcing simulation

Phase I will gather and organize the minimum set of farm and buyer inputs necessary to run a capacity-banded coordination model for the Summit County study area.

Output: populated test dataset and scenario model

Success condition: enough valid inputs exist to run bounded scenario sweeps and compare outcomes against a defined baseline

Objective 4: Determine whether a two-sided feasibility window exists

Phase I will evaluate whether any tested scenario yields an outcome in which participating farms improve their position while buyers remain at least cost-neutral or otherwise non-worse-off within the defined thresholds.

Output: feasibility assessment report

Success condition: the analysis identifies either a credible feasible window or a clear explanation of why none exists within the tested conditions

VI. Phase I Research Design and Work Plan

This section is the operational core of the proposal. The study is best described as a **bounded feasibility investigation using a capacity-banded coordination model rather than a full agronomic optimization model**. Its purpose is not to maximize biological yield. Its purpose is to determine whether realistic sourcing scenarios can be identified under observed production ranges and bounded assumptions.

A. Governing feasibility claim

Phase I will test whether it is feasible to use an open-source coordination framework to identify local sourcing scenarios in Summit County, Ohio, in which participating farms improve their stability or expected gain while grocery or hub buyers remain at least non-worse-off under defined cost and service conditions.

B. Decision rule

The project will be treated as **feasible for Phase II advancement** if the study identifies at least one credible scenario in which both producer-side and buyer-side conditions are satisfied within the defined geographic, seasonal, and handling constraints.

The project will be treated as **not feasible under current conditions** if no tested scenario satisfies both sides of that requirement, or if the conditions required to satisfy it fall outside the study's bounded assumptions.

C. Core research questions

1. What exact uncertainty is being tested?

Whether coordination rather than demand absence explains some portion of current local agricultural underutilization.

2. By what method is it being tested?

By building a minimum coordination framework, collecting bounded farm and buyer inputs, and running scenario-based feasibility analysis.

3. Against what baseline is it being tested?

Against current local fragmentation and a bounded outside-supply comparator, both defined explicitly in the final proposal.

4. What result justifies Phase II?

Identification of a two-sided feasible window in which farms improve their position while buyers remain non-worse-off.

D. Minimum Phase I artifacts

Phase I will produce only the minimum artifacts required to test the governing claim:

• **Artifact 1: Structured input model**

A minimum data layer for farm-side, buyer-side, and scenario-side variables.

• **Artifact 2: Prototype coordination engine**

A prototype that can accept inputs, normalize them, apply bounded constraints, and generate interpretable outcomes.

• **Artifact 3: Feasibility simulation**

A capacity-banded scenario model that tests the presence or absence of feasible sourcing windows.

• **Artifact 4: Decision report**

A study output that identifies whether Phase II is warranted, and why.

E. Study boundaries

The study is geographically bounded around Summit County, Ohio, with local or regional sourcing assumptions defined explicitly in the final proposal. It is focused on small and independent farm participation in local or regional supply systems. It does not attempt to prove national scalability, universal local competitiveness, or a complete deployment of the broader framework within Phase I.

F. Fixed versus variable elements

To keep the study bounded, some elements will be treated as fixed and others as scenario variables.

Fixed for Phase I:

- geographic study boundary
- general buyer class in scope
- bounded comparator assumptions
- study timeframe
- minimum product basket assumptions

Variable within the model:

- coordination level
- farm participation level
- product availability overlap
- adjacency and route conditions
- waste or shrink assumptions
- buyer-side tolerance conditions

G. Minimum data inputs

The study will collect only the data needed to run the feasibility test. These categories are expected to include:

Farm-side inputs

- current output bands
- acreage and capacity bands
- harvest windows or timing bands
- location and adjacency factors
- handling, storage, or processing constraints
- other operational limits relevant to the model

Buyer-side inputs

- bounded demand profile
- product basket assumptions
- frequency or timing expectations
- service or procurement requirements
- delivered-cost tolerance or other bounded threshold

Scenario-side inputs

- coordination assumptions
- routing or distance assumptions
- waste or shrink assumptions
- aggregation assumptions

Table 1

Minimum Phase I data fields by farm, buyer, and scenario type.

H. Data trust basis

The final proposal will state the basis on which Phase I inputs are considered credible enough for feasibility testing. These inputs may include structured interviews, current operating records, planning information, historical observations, and bounded scenario assumptions. The proposal will not claim more precision than the input quality allows.

I. Modeling approach

The model will treat current output and capacity as **bands**, not point estimates. This is deliberate. The study is designed to evaluate transitional feasibility under uncertainty, not to overstate the precision of future production. The model will test whether realistic sourcing combinations exist under bounded assumptions involving transportation, timing, variety overlap,

handling limits, and coordination level. National evidence supports the use of capacity bands for fresh produce: USDA Economic Research Service data document total losses of 37 percent for fresh fruit and 34 percent for fresh vegetables at the retail and consumer levels, confirming that timing, handling precision, and logistics assumptions are material drivers of outcome variation in perishable-produce coordination models (USDA ERS, 2025). Hub-network modeling further demonstrates that economies of scale affect aggregation feasibility across farm size, geography, and seasonal variation, which is why the study evaluates scenario ranges rather than single-point assumptions (Ge et al., 2019).

The purpose of the model is to reveal whether a feasible coordination window exists. It is not intended to serve as a definitive agronomic forecast or as proof that all farms in the study area can or should supply all relevant demand.

J. Evaluation logic

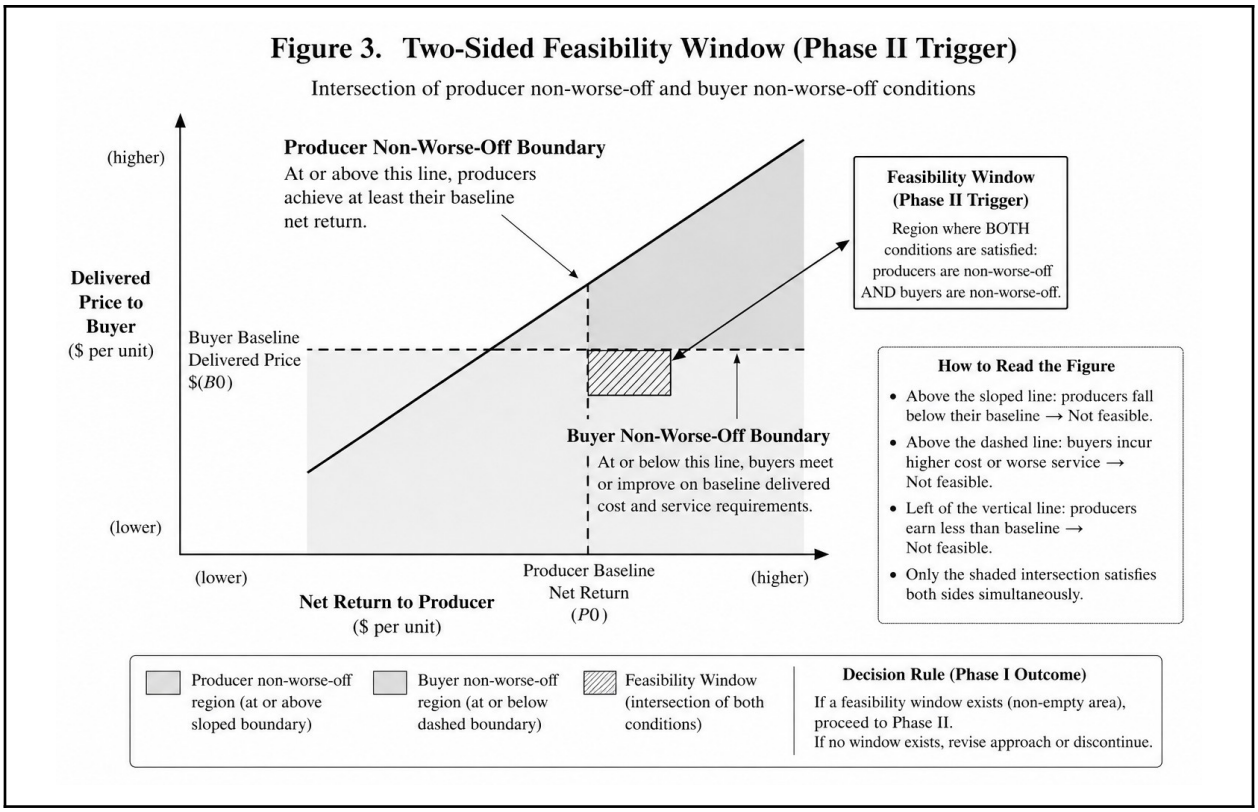
The study will evaluate whether there is an intersection between:

• a **producer-side condition**, where farm stability or expected gain improves or at minimum does not worsen relative to the chosen baseline; and

• a **buyer-side condition**, where delivered cost, service, or other selected threshold remains at least non-worse-off relative to the chosen comparator.

That intersection will be treated as the **two-sided feasibility window**. This evaluation structure is consistent with opportunity-cost frameworks in the regional food-systems literature, which require that positive local-system effects be assessed net of what is displaced elsewhere, not merely as one-sided producer gains (Jablonski et al., 2016).

Figure 3



Two-sided feasibility window showing producer-side and buyer-side non-worse-off intersection.

K. Eight-month task sequence

Task 1 – Scope definition and minimum data specification (Months 1–2)

Finalize study boundaries, user classes, minimum variables, and data requirements. Translate the study question into a structured input model and finalize the minimum schema requirements for Phase I.

Deliverables:

- finalized feasibility-question statement
- minimum variable list
- draft structured data specification

Task 2 – Prototype coordination framework development (Months 2–4)

Develop the minimum prototype needed to accept, normalize, and process farm-side and buyer-side inputs for scenario evaluation.

Deliverables:

- working prototype for study use
- input/output structure for scenario testing

Task 3 – Partner input collection and model parameterization (Months 4–6)

Gather bounded farm and buyer inputs, validate that they are sufficient for the study, and populate the Phase I scenario environment.

Deliverables:

- populated bounded test dataset
- documented assumptions and scenario variables

Task 4 – Scenario analysis and feasibility-window identification (Months 6–7)

Run scenario sweeps across the defined variable conditions to determine whether a two-sided feasible window exists and which constraints dominate infeasibility when it does not.

Deliverables:

- scenario comparison outputs
- preliminary feasibility finding

Task 5 – Phase I synthesis and Phase II transition logic (Months 7–8)

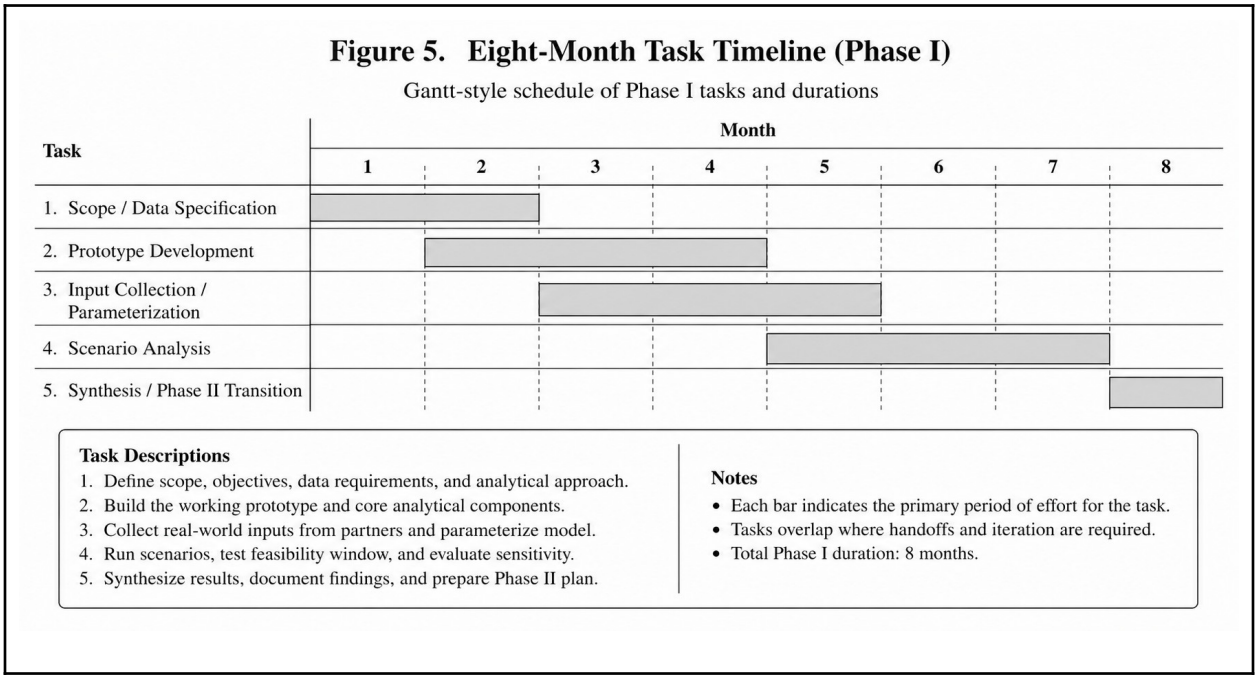
Synthesize findings into a final feasibility assessment and define what would be required to operationalize the model in Phase II if feasibility is supported.

Deliverables:

- Phase I feasibility report
- Phase II transition requirements

• summary of limiting constraints

Figure 4



Gantt-style eight-month timeline.

VII. Related Research and Competition

This project sits in relation to, but is distinct from, several categories of existing approaches:

• proprietary farm-management systems

• digital marketplace or food-hub platforms

• recordkeeping and compliance tools

• centralized coordination models dependent on one intermediary

• software that improves internal operations without enabling interoperable coordination across independent actors

The project's claim is not that these tools have no value. It is that they often do not solve the specific problem being tested here: whether multiple independent farms and bounded buyer

demand can be represented and evaluated through a shared coordination layer without requiring centralized ownership of the network. The empirical literature provides additional grounds for disciplining market assumptions. Consumer willingness to pay for local food exhibits substantial heterogeneity and is not reliably large enough to absorb coordination or delivery cost premiums: a meta-regression analysis found evidence of publication bias in the willingness-to-pay literature (Printezis et al., 2019), and a field experiment found no statistically significant premium from a generic local label under competitive retail conditions (Davidson et al., 2024). Persistent buyer-side barriers—including price competitiveness, delivery reliability, processing and packaging fit, and procurement procedure compatibility—remain the primary constraints on institutional market expansion for local and regional supply (Bielaczyc et al., 2020; Barry et al., 2025; Qi et al., 2017). These findings collectively support the proposal's decision to anchor the feasibility test in operational cost-and-service metrics rather than in assumed consumer preference.

Prior work

Fruitful Network Development’s prior work includes software and data-organization activity oriented toward small agricultural operations, including website hosting, data organization, and operational tooling. This prior work is relevant because it places the proposed study in continuity with actual user-facing development rather than pure theory.

Distinguishing contribution

The distinguishing contribution proposed here is a schema-first, open-source coordination approach. Rather than building value only inside a closed application or marketplace, the project aims to test whether interoperability itself can become a meaningful coordination primitive for local agricultural use.

VIII. Market Opportunity and Commercialization Path

The national evidence base establishes a meaningful but structurally constrained local food market. The 2020 Local Food Marketing Practices Survey reported $9.0 billion in direct farm sales of local edible food, with intermediaries and institutions accounting for 46 percent of those sales and direct-to-consumer channels for only 33 percent (Dorn, 2022). This channel distribution confirms that scalable local food commercialization depends on satisfying institutional and intermediated buyer requirements—including price competitiveness, delivery reliability, and audit compliance—not merely on consumer preference alone. The commercial opportunity is therefore in reducing the coordination friction that currently prevents local supply from satisfying buyer reliability standards rather than in capturing a straightforward consumer premium. At the structural level, most food-system value accrues after the farm gate: farmers received 15.9 cents of each dollar spent on domestically produced food in 2023, with the remaining share distributed across processing, transportation, wholesaling, retailing, and related functions (USDA ERS, 2026). A viable commercialization path must demonstrate logistics efficiency and service-level reliability at the coordination layer, not only farm-level production potential.

Phase I is not intended to commercialize a final system. It is intended to determine whether the technical and economic conditions for commercialization are present. If Phase I identifies a feasible window, Phase II would move from analysis toward operational deployment. That later stage could include broader onboarding, recurring data flows, more active coordination workflows, and clearer productization of the framework for farm and buyer use.

The early commercialization logic is therefore conditional:

• **If Phase I is positive:** operationalize the validated minimum commitments and workflows.

• **If Phase I is negative:** refine the model, change the study boundary, or suspend the implementation path until the limiting constraints are better understood.

Target user classes

Primary user classes are expected to include small and independent farms that need structured planning and coordination support. Secondary user classes may include local food hubs, grocers, cooperatives, and other bounded buyer or intermediary entities.

Business-model direction

The business-model direction remains open at the level of exact final packaging, but the project is oriented toward an open-source core with practical service, tooling, and implementation layers that do not depend on monopolizing the underlying network.

IX. Resources, Facilities, and Team

Fruitful Network Development LLC is positioned to conduct this Phase I work as a small, technically focused organization with a bounded and realistic scope.

Company and project readiness

The company develops software and data-schema infrastructure for decentralized coordination across agricultural operations. Its focus is on interoperable data models, farm-level tooling, and system architectures intended to improve the visibility of supply, demand, and logistics while preserving local autonomy.

Principal Investigator qualifications

The Principal Investigator has developed this technology over multiple years and has pursued it through technical, entrepreneurial, and field-oriented iteration. Relevant background includes computer engineering and applied mathematics training at the University of Akron, direct software development experience, and a sustained effort to move the concept from

hypothesis to working framework. The PI has also completed extensive market and customer discovery work related to the agricultural problem space.

Execution qualifications

The project is not being proposed from a standing start. The PI has already refined the concept through customer discovery, iteration, and direct interaction with local agricultural networks. Existing relationships with farms, community-serving agricultural organizations, and regional innovation-support structures provide a credible base for bounded Phase I collaboration and input gathering.

Facilities and equipment

The facilities required for Phase I are modest and consistent with the scope of the work. The project requires secure development workspace, local workstations adequate for software development, cloud infrastructure sufficient to host and test the prototype environment, and access to collaborating farms or local stakeholders for bounded input collection. The proposal does not depend on large capital equipment purchases.

Administrative capability

The company's Phase I structure is intended to remain focused and manageable. The proposal assumes the ability to track work time, organize project records, manage development tasks, and document expenses in a manner appropriate to a small federally funded research effort. The resources requested in the final proposal will be tied directly to the technical objectives and deliverables of the study.

X. Anticipated Significance

The significance of the proposed work lies in its disciplined treatment of a problem that is often discussed too broadly. Rather than claiming to solve local agriculture in general, this

proposal isolates one actionable uncertainty: whether improved coordination and legibility can create a feasible transition pathway under bounded conditions (Key et al., 2017; Bielaczyc et al., 2020; Ge et al., 2019).

A positive result would matter because it would show that some portion of current underutilization is not merely an unavoidable economic fact, but a condition shaped by information, planning, and coordination friction that can be tested and reduced. A negative result would also matter because it would clarify which constraints remain binding and would prevent later development from being pursued on the basis of vague optimism.

This proposal therefore treats Phase I as a decision-oriented technical study. Its value lies in making the next step rational, whether that next step is implementation, revision, or suspension of the current path.

XI. Conclusion

This narrative proposes a bounded Phase I study intended to test whether an open-source coordination framework can identify realistic local sourcing scenarios in Summit County, Ohio, in which participating farms improve their position while buyers remain at least non-worse-off.

The proposal is structurally a planning proposal, but the work it advances is a research-style feasibility investigation. That hybrid is deliberate. The plan being proposed is not immediate deployment of a full system. It is a disciplined test of whether deployment is justified.

The proposal therefore supports one decision: whether USDA should support an eight-month Phase I effort that defines minimum data structures, develops a bounded prototype, parameterizes a local scenario model, and determines whether a two-sided feasibility window exists. By keeping the scope narrow, the method explicit, and the outputs decision-oriented, the

project aligns with the purpose of a technical academic proposal and creates a clearer threshold for Phase II advancement.

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